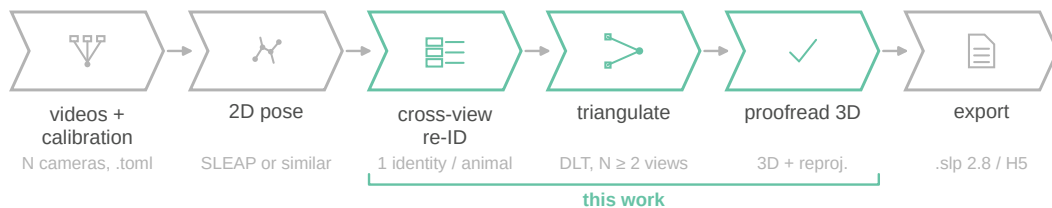
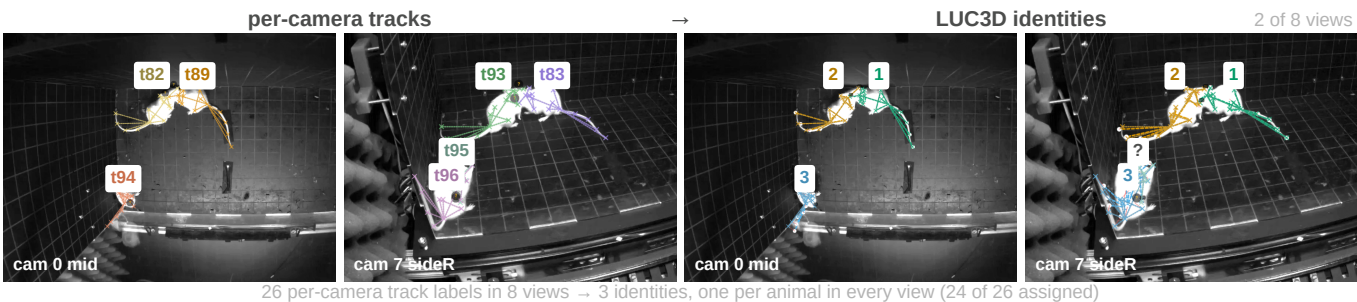


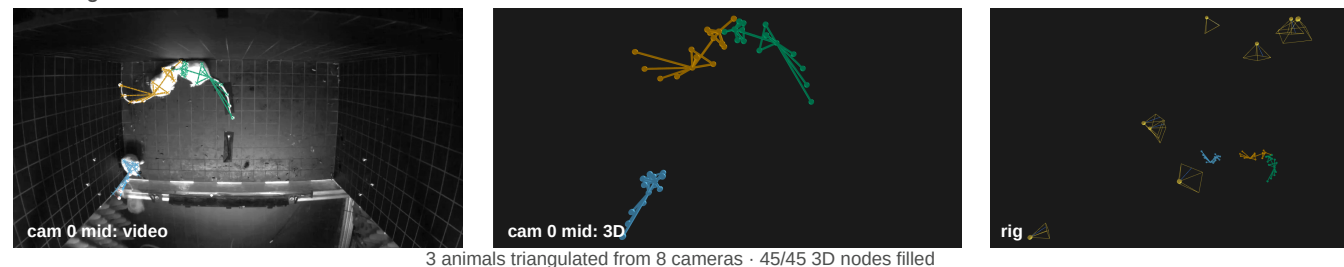
a Pipeline



b Cross-view re-identification



c Triangulated 3D



d Capability comparison

	Install	Runs in	Multi-animal	Multi-camera 3D	Cross-view ID	3D proofreading
LUC3D (this work)	none	browser	✓	✓	✓	✓
SLEAP	conda/pip	desktop	✓	sleap-anipose	—	2D only
Label3D	MATLAB	desktop	1 animal	✓	—	✓
JARVIS (HybridNet)	conda/pip	desktop	1 animal	✓	—	reproj. error
Lightning Pose	pip	browser + CLI	1 animal	multi-view losses	—	—
DeepLabCut	conda/pip	desktop	✓	1 animal, pairwise	—	—
Anipose	conda/pip	CLI	—	✓	—	—
DANNCE / SDANNCE	conda/pip	CLI + MATLAB	✓	✓	—	—

third-party cells from published docs, checked 2026-08-04 — re-verify before submission

a: schematic. b, c: one frame of an 8-camera HardFight recording, driven through LUC3D itself (load, Track All, Triangulate All).

d: third-party capabilities from published documentation, checked 2026-08-04; qualifications in the caption.